

ICT 3105

AI and Expert Systems

Marks: 20

Time: 30 min

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| 1. Write some elements of a production system. | 5 |
| 2. Give an example of an OR graph. | 5 |
| 3. Write (i) Union, (ii) Intersection and (iii) Complement for Fuzzy sets. | 5 |
| 4. What do you understand by the term 'State Space' in the AI domain? | 5 |

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|---|---|
| 1. State the Bayes theorem. | 2 |
| Let $P(A) = 0.10$, $P(B) = 0.050$ and $P(B A) = 0.070$. What is the value of $P(A B)$? | 4 |
| 2. What is the Certainty factor? | 2 |
| How does one distinguish between measures of disbelieve and measure of believe? | 4 |
| 3. How does one distinguish between monotonic and nonmonotonic reasonings? | 4 |
| 4. Write in logic form: If it is not cold, we shall go for swimming. | 4 |

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|---|---|
| 1. Name the steps in the process in the natural language understanding process. | 5 |
| 2. How would it look like in top-down and bottom-up parsing: 'the girl ate a burger'. | 5 |
| 3. Name the Chomsky hierarchy of generative grammars. | 5 |
| 4. Explain in game playing format: 'a block is placed on another block'. | 5 |

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CT 03

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|--|---|
| 1. Why is Alpha-Beta pruning better than Min-Max search procedure? | 5 |
| 2. Name five Expert Systems. | 5 |
| 3. Make an association network for learning from the sentence: I am Delta. I'm a student. | 5 |
| 4. Mention the four major factors that contribute to the difficulty of learning. | 5 |